

SHELL SCRIPTING PROGRAMS WITH OUTPUT

1. Arithmetic Operations Using expr

```
#!/bin/bash
read a b

add=`expr $a + $b`
sub=`expr $a - $b`
mult=`expr $a \* $b`
div=`expr $a / $b`

echo "$add"
echo "$sub"
echo "$mult"
echo "$div"
```

Sample Output:

```
Input:
10 5

Output:
15
5
50
2
```

2. Check Leap Year or Not

```
#!/bin/bash
read num

if [ $((num % 4)) -eq 0 ]
then
    echo "leap year"
else
    echo "not a leap year"
fi
```

Sample Output:

```
Input:
2024

Output:
leap year
```

3. Reverse a Number

```
#!/bin/bash

echo "Enter a number:"
read num

rev=0

while [ $num -gt 0 ]
do
    rem=$((num % 10))
    rev=$((rev * 10 + rem))
    num=$((num / 10))
done

echo "Reversed number = $rev"
```

Sample Output:

```
Input:
1234

Output:
Reversed number = 4321
```

4. Fibonacci Series

```
#!/bin/bash
echo "Enter number of terms:"
read n
a=0
b=1

echo "Fibonacci series:"
echo "$a"
echo "$b"

for (( i=3; i<=n; i++ ))
do
    c=$((a + b))
    echo "$c"
    a=$b
    b=$c
done
```

Sample Output:

```
Input:
5

Output:
0
1
1
2
3
```

5. CPU Utilization

```
#!/bin/bash
Mem=`df -h . | tail -1 | awk -F " " '{print $4}' | sed 's/%//g'`

if [ $Mem -gt 40 ]
then
    mail -s "memory reached" user@gmail.com < filename
fi
```

Sample Output:

```
Output:  
Mail sent when memory usage exceeds limit
```